

NUMBER PLATE RECOGNITION SYSTEM

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology **in** **Electrical and Electronics** **&** **Electronics and Instrumentaion**

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May, 2025.

DECLARATION

We here by declare that the thesis entitled “**NUMBER PLATE RECOGNITION SYSTEM**” submitted by Swastik Panigrahi, Shivank Mehra and Oindrila Choudhury for the award of the degree of *Bachelor of Technology in Electrical and Electronics Engineering* to VIT University is a record of bonafide work carried out by us under the supervision of **Dr. Karthikeyan A**, Associate Professor Sr, Select School, VIT University, Vellore.

We further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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This is to certify that the thesis entitled “**NUMBER PLATE RECOGNITION SYSTEM**” submitted by **Swastik Panigrahi(21BEE0145), Oindrila Choudhury(21BEE0248), Shivank Mehra(21BEI0056)**, School of Electrical Engineering, Vellore Institute of Technology, Vellore, for the award of *Bachelor of Technology in Electrical and Electronics Engineering* , is a record of bonafide work carried out by them under my supervision during the period, 13.12.2024 to 18.04.2025, as per the VIT code of academic and research ethics.

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Executive Summary

Anyone who's circled a crowded parking lot knows the frustration of urban parking. This research tackles the problem head-on with a smart system that automatically reads license plates and guides drivers to open spots. Using regular security cameras combined with simple sensors, the system recognizes arriving cars, tracks how long they park, and handles payments automatically - no more paper tickets or struggling with payment machines. The real magic happens in how it processes images. While many systems struggle with blurry or angled license plates, this one combines clever edge-detection techniques with Google's powerful image recognition to achieve near-perfect accuracy. The researchers acknowledge some limitations - very dirty plates or extreme angles can still cause issues - but future upgrades might include mobile app integration and support for electric vehicle charging stations. This isn't just theoretical - the system represents a realistic, cost-effective way to modernize parking infrastructure. As cities grow more crowded, such smart parking solutions could save drivers countless hours while helping businesses and municipalities optimize their parking assets.

Meanwhile, sensors track exactly how long you've parked, so you only pay for the time you actually used. The brains behind this system combine clever camera technology with simple but effective sensors to make parking hassle-free. What's really impressive is how it handles tricky situations - whether it's bad lighting, odd angles, or weathered license plates, the system gets it right 98 times out of 100. That's better than most human attendants! While it's not perfect (heavily obscured plates can still stump it), the researchers are already working on upgrades like mobile app integration and support for electric vehicle charging. This isn't just lab research - it's a practical solution that could soon make circling for parking spots and overpaying for parking a thing of the past. As cities get more crowded, smart systems like this could save everyone time, money, and a whole lot of frustration.

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LIST OF TERMS AND ABBREVIATIONS

3.1	ANPR	Automatic Number Plate Recognition	1
4.1	OCR	Optical CharacterRecognition	1
5.1	LPR	License Plate Recognition.....	1
6.1	ML	Machine Learning.....	1
7.1	IP	Internet Protocol	3,6

CHAPTER 1

INTRODUCTION

1.1 Objective

This study aims to address the urban vehicle parking issues by proposing a solution using Automatic Number Plate Recognition (ANPR) through image processing and a sensor-based hardware system. Integrating these technologies forms a Smart Parking Management System (SPMS) to automate parking processes and enhance the parking experience. The study aims to create an efficient system that eliminates manual vehicle registration and optimizes space utilization. ANPR and IoT-based sensors help users identify the available slots and pay only for the actual parking duration, which will help to minimize the fixed billing rates. The proposed ANPR system processes vehicle number plates at entry, ensuring seamless identification and eliminating manual registration. IoT sensors monitor real-time slot occupancy, transmitting data to a web admin panel. This panel provides insights such as entry and exit times, total parking duration, and billing costs, facilitating efficient management and remote monitoring. The ANPR-based SPMS reduces reliance on manual processes, streamlining entry procedures. By dynamically assessing slot availability through IoT sensors, users can locate unoccupied spaces quickly, which enhances user convenience. The web admin panel allows administrators to monitor the system remotely, ensuring smooth operations and maintaining accurate records. This study introduces a comprehensive solution to urban parking challenges by integrating ANPR and IoT technologies. The SPMS improves efficiency, reduces human resource needs, and enhances user experience with flexible billing based on actual duration. The combination of hardware and software provides a foundation for effective urban parking management.

1.2 Motivation

With cities getting smarter and the need for better traffic control and security growing every day, number plate recognition systems have become more important than ever. While there's already been a lot of work done in this area, many existing systems are either too expensive, don't offer real-time updates, or aren't very flexible.

That's what got me interested in this project. I wanted to explore how IoT—especially

using simple, affordable tools like the ESP32-CAM—can help create a smarter, more efficient number plate recognition system. My goal is to design something that can automatically capture and send license plate data in real-time, making it useful for things like parking management or traffic monitoring.

This project gives me a chance to combine what I've learned about IoT, computer vision, and cloud tech while working on something that could actually help improve everyday life.

1.3 Background

Managing parking efficiently has become a major concern in cities due to the rising number of vehicles. Traditional parking systems often involve manual processes, which can lead to wasted time, congestion, and inefficiencies. That's where smart technologies like Automatic Number Plate Recognition (ANPR) and the Internet of Things (IoT) come in. ANPR systems use cameras and image processing to automatically read vehicle license plates, while IoT devices like sensors help monitor real-time parking availability.

This combination of ANPR and IoT makes it possible to create intelligent parking systems that can guide drivers to free spots, log entry and exit times, and even automate billing—all without human intervention. With hardware like ESP32 microcontrollers and tools like Google Cloud Vision for OCR, it's now easier and more affordable than ever to build these smart systems. The shift toward these technologies is part of a larger move to make urban living smarter, more efficient, and more convenient for everyone.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 Review of Literature

Intelligent parking systems have gained significant attention due to the rapid increase in vehicle numbers and the need for efficient parking management. Automated License Plate Recognition (ALPR) plays a crucial role by enabling quick and accurate vehicle identification without human intervention. To enhance security, modern systems like the one proposed by Van et al. (2022) integrate ALPR with face verification of drivers. This dual-authentication method ensures that only authorized users can access parking facilities. Such innovations contribute to the development of smarter, more secure urban infrastructure. (Sharma et al. 2022)

Van et al. (2022) proposed an intelligent parking system combining Automated License Plate Recognition (ALPR) and face verification to enhance security and efficiency. ALPR identifies vehicles quickly, while face verification ensures the driver's authenticity. Their system addresses challenges like unauthorized access and improves real-time parking management. Compared to earlier models, this dual-authentication approach provides higher reliability. The study contributes to advancing smart parking solutions within modern urban environments. (Van et al. 2022)

Kashyap et al. (2018) presented a system for Automatic Number Plate Recognition (ANPR) aimed at improving vehicle identification accuracy. Their approach focuses on efficient image processing techniques to detect and recognize license plates under various conditions. The study highlights challenges like lighting variations, plate damage, and font differences. By enhancing recognition reliability, their system supports applications in traffic management and security. This research contributes to the growing field of intelligent transportation systems. (Kashyap et al. 2018)

Farooqi et al. (2019) developed UParking, a smart parking management system leveraging the Internet of Things (IoT) to optimize parking space usage. Their system uses

sensors and real-time data to monitor and guide vehicles efficiently. The research addresses common parking issues like space shortages and traffic congestion. By integrating IoT technologies, UParking improves accessibility, convenience, and urban traffic flow. This work contributes to advancing smart city infrastructure through intelligent parking solutions. (Farooqi et al. 2019)

Thakre et al. (2021) proposed an IoT-based smart vehicle parking system utilizing RFID technology for automated vehicle identification. Their system aims to reduce manual intervention and streamline the entry and exit processes in parking lots. By using RFID tags, vehicles are quickly detected, leading to faster and more secure parking management. The study highlights the system's efficiency in reducing congestion and improving user convenience. Compared to traditional systems, this approach offers better scalability and automation. Their work supports the growing shift toward intelligent and connected urban infrastructures. Thakre et al. (2021)

Ahmed et al. (2019) proposed a blockchain-based architecture to enhance the security and transparency of integrated smart parking systems. Their model ensures decentralized management of parking data, reducing risks of data tampering and unauthorized access. The system supports real-time parking information, user authentication, and secure payments. By combining IoT sensors with blockchain technology, it offers a more reliable and scalable parking solution. The study addresses limitations of traditional centralized systems. Their work paves the way for future smart city developments integrating blockchain with IoT. (Ahmed et al. 2019)

Pasala et al. (2022) introduced the Smart Parking System (SPS), an intelligent parking solution based on advanced image processing techniques. Their system detects available parking spots by analyzing real-time visual data, reducing the need for manual monitoring. It improves parking efficiency by providing drivers with accurate, up-to-date information on vacant spaces. The study addresses challenges such as varying lighting conditions and vehicle occlusions. Compared to sensor-based methods, their image-based approach is cost-effective and scalable. This work supports smarter urban mobility through vision-driven technologies.(Pasala et al. 2022)

Singh and Bhushan (2019) proposed a real-time Indian license plate detection system using deep neural networks (DNNs) combined with Optical Character Recognition (OCR) through Long Short-Term Memory (LSTM) and Tesseract. Their system improves accuracy in detecting and recognizing license plates under diverse environmental conditions. The integration of LSTM with Tesseract enhances the recognition of characters despite variations in fonts and plate designs. The study addresses challenges like

noise, poor image quality, and occlusions. Compared to traditional methods, their approach achieves higher accuracy and real-time performance. This research contributes to advancing vehicle identification systems for traffic management and security. (Singh and Bhushan 2019)

Kashyap et al. (2018) presented an Automatic Number Plate Recognition (ANPR) system aimed at improving vehicle identification for various applications such as traffic monitoring and security. Their approach uses image processing techniques to efficiently detect and recognize number plates. The system focuses on enhancing recognition accuracy under varying environmental conditions, including lighting and plate design differences. It incorporates advanced algorithms to address common challenges like plate occlusion and image noise. The study offers a robust solution for real-time vehicle identification. This research contributes to the development of intelligent transportation systems and automated vehicle management. (?)

Hsieh et al. (2022) introduced an Automatic Vehicle License Plate Recognition (LPR) system based on YOLO v4 for smart parking management. Their approach leverages the power of deep learning, specifically the YOLO v4 model, to accurately detect and recognize vehicle plates in real-time. The system addresses key challenges such as high vehicle density and varying lighting conditions. YOLO v4's efficiency allows for faster processing and better accuracy compared to traditional methods. This solution contributes to the development of smarter and more automated parking systems. The study demonstrates YOLO v4's potential in enhancing urban parking infrastructure. (Hsieh et al. 2022)

Sheng et al. (2023) proposed a parking system based on Radio-Frequency Identification (RFID) technology to automate vehicle identification and improve parking management. Their system uses RFID tags for quick detection, enabling efficient space allocation and minimizing human intervention. The approach enhances parking security by accurately tracking vehicle entry and exit in real-time. The study highlights RFID's potential in addressing common parking challenges such as congestion and unauthorized access. Compared to traditional systems, RFID offers a more cost-effective and scalable solution. This work contributes to the development of intelligent parking systems in modern urban settings. (Sheng et al. 2023)

2.2 Project Description

This project focuses on developing a smart parking management system using Automatic Number Plate Recognition (ANPR) and Internet of Things (IoT) technologies. The idea is to solve common problems in urban parking—like long waiting times, inef-

efficient space usage, and the hassle of manual billing—by automating the entire process.

The system uses ESP32 microcontrollers, infrared (IR) sensors, and cameras to monitor and manage parking slots in real time. When a vehicle arrives at the entrance, an IR sensor detects it and triggers a camera to capture the number plate. The image is then processed using Google Cloud Vision AI, which extracts the plate number with high accuracy. This information is stored in a database along with the timestamp.

If a parking slot is available, the system automatically opens the entrance gate. While the vehicle is parked, IR sensors monitor which slots are occupied and send updates to the microcontroller, which then reflects the current status on an LCD display at the entrance. When the vehicle exits, a similar process happens at the exit gate—the number plate is re-captured, exit time is recorded, and the total parking duration is calculated automatically. Based on the duration, the system generates a bill and displays it to the user.

All this data—entry and exit times, available slots, vehicle details, and billing—is managed through a web-based dashboard, giving administrators complete control and real-time insights from anywhere.

This project not only helps in reducing human involvement and errors but also provides a more efficient and user-friendly parking experience. It's a step forward toward building smarter and more sustainable cities.

2.2.1 Software

The software side of this smart parking system brings together a mix of programming languages, tools, and platforms to make the whole system work smoothly—from capturing number plates to managing parking slots and displaying data on a dashboard.

Here's what's used:

1. Embedded C Used to program the ESP32 microcontroller, which controls sensors, LCD displays, and gate mechanisms. This handles all the hardware-level interactions.

2. Python Python scripts are responsible for:

- Capturing and processing the number plate images.

- Using Google Cloud Vision API for Optical Character Recognition (OCR).

- Connecting to the database using an SQL connector library.

- Running logic for entrance/exit detection and billing calculations.

3. Google Cloud Vision AI This powerful cloud-based OCR tool is used to read and extract text from the captured number plate images. It offers much better accuracy than traditional OCR engines like Tesseract.

4. PHP PHP is used as the middleware between the ESP32 and the database. It handles:

- CRUD operations (Create, Read, Update, Delete) in the MySQL database.

Communication between the web server and the ESP32 using HTTP requests.

5. HTML, CSS, and JavaScript These front-end web technologies are used to create a user-friendly dashboard for admins to view:

- Current parking slot status.

- Vehicle entry and exit logs.

- Billing information.

6. MySQL A relational database used to store all system data including:

- Vehicle number plates.

- Entry and exit times.

- Slot availability and billing details.

7. OpenCV (Computer Vision Library) Used in Python to pre-process images (e.g., grayscale conversion, edge detection using Canny algorithm) before sending them to the OCR engine.

This combination of tools makes the system responsive, accurate, and easy to manage in real time—whether you’re a developer working behind the scenes or an administrator monitoring the parking lot.

2.2.2 Hardware

The proposed Automatic Number Plate Recognition (ANPR) integrates a combination of hardware and IoT technologies to automate parking processes. The core hardware components include:

- ESP32 Microcontroller:** Acts as the central control unit, interfacing with sensors, cameras, and actuators. It connects to Wi-Fi to transmit and receive data between the PHP script and MySQL database, enabling real-time updates and remote management.

- Infrared (IR) Sensors:** Deployed at entry, exit, and individual parking slots to detect vehicle presence and monitor slot occupancy. These sensors send signals to the ESP32, which updates the system database to reflect real-time availability.

- HD Cameras:** Positioned at entry and exit points to capture vehicle license plates. These cameras work in tandem with image processing algorithms (e.g., Canny edge detection) and Google Vision AI Cloud OCR to extract and recognize plate numbers accurately.

- Servo Motors:** Control the entry and exit barrier gates, automatically opening and closing them based on signals from the ESP32 after successful license plate recognition.

- Liquid Crystal Displays (LCDs):** Located at entry and exit points to display parking slot availability, timestamps, and billing information to users. The entrance LCD also alerts drivers when all slots are occupied.

- Power Supply:** Utilizes an 18650 battery for portable operation, ensuring uninterrupted functionality of the microcontroller and connected peripherals.

- Computer/Laptop:** Hosts the Python-based image processing and OCR software,

managing license plate detection, data comparison, and billing calculations. It interfaces with the ESP32 via Wi-Fi to synchronize hardware actions with backend databases.

The hardware architecture is designed for scalability and efficiency, minimizing manual intervention while optimizing parking space utilization. Figure 3 in the paper illustrates the schematic layout of these components, highlighting their interconnected roles in the system.

2.3 Goals

The primary goal of this research is to develop an Automatic Number Plate Recognition (ANPR) that addresses urban parking challenges by leveraging IoT, image processing, and real-time data analytics. Key objectives include:

Automation of Parking Processes – Eliminate manual vehicle registration and billing by using ANPR technology to seamlessly identify vehicles at entry and exit points, reducing waiting times and human errors.

Optimized Space Utilization – Employ IoT-based infrared (IR) sensors to monitor parking slot occupancy in real time, enabling drivers to quickly locate available spaces and improving overall parking efficiency.

Dynamic Billing System – Calculate parking fees based on actual duration of stay, ensuring fair pricing and minimizing fixed-rate inefficiencies.

Remote Management Data Analytics – Provide administrators with a web-based dashboard to monitor parking status, entry/exit logs, and financial records, facilitating better decision-making and system oversight.

Enhanced User Experience – Reduce congestion and search time for drivers while ensuring a secure, contactless parking process through automated barriers and real-time slot availability displays.

Scalability Integration – Design a modular system that can be adapted for various parking facilities and integrated with smart city infrastructure, including future expansions like electric vehicle charging stations and mobile app reservations.

By combining ANPR, IoT sensors, and cloud-based OCR, this system aims to transform traditional parking into an intelligent, efficient, and user-friendly solution for urban environments.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 Software Specification

The proposed system relies on a combination of software frameworks, libraries, and programming languages to enable seamless license plate recognition, real-time data processing, and system automation. Below are the key software components:

1. Core Programming Languages Frameworks Embedded C – Used for programming the ESP32 microcontroller to interface with sensors, servo motors, and LCD displays.

Python – Handles image processing, ANPR, and data management through OpenCV, Tesseract OCR (for testing), and Google Vision AI Cloud (for production).

Web Development (HTML, CSS, PHP, JavaScript) – Powers the admin dashboard, enabling real-time monitoring of parking slots, vehicle logs, and billing records.

SQL (MySQL Database) – Stores vehicle entry/exit records, parking durations, billing details, and slot availability.

2. Image Processing Optical Character Recognition (OCR) OpenCV – Used for license plate detection, image filtering (grayscale, blur), and edge detection (Canny algorithm).

Google Vision AI Cloud – Provides high-accuracy OCR (98

Tesseract OCR (for testing) – Evaluated but found less accurate than Google Vision AI for real-world conditions.

3. Real-Time Data Processing Communication Wi-Fi (ESP32 Connectivity) – Enables communication between hardware (sensors, cameras) and the central database.

PHP Scripts – Facilitate CRUD (Create, Read, Update, Delete) operations on the MySQL database, ensuring seamless data flow between ESP32 and the backend.

SQL Connector (Python Library) – Allows Python scripts to query the database for real-time updates on vehicle entries/exits.

4. System Workflow Automation Entry Process:

IR sensor detects a vehicle → triggers camera capture.

Python script processes the image (OpenCV + Google Vision AI) → extracts license plate.

ESP32 updates the database and opens the gate.

Exit Process:

Repeats ANPR → matches entry record → calculates parking duration bill.

Displays details on LCD → opens exit gate after payment confirmation.

Admin Dashboard:

Provides real-time parking slot status, vehicle logs, and billing reports.

Supports filtering by date, vehicle number, or duration.

5. Future Software Enhancements Mobile App Integration – Allow users to reserve slots, make payments, and receive notifications.

Blockchain for Secure Transactions – Enhance data integrity for billing and user privacy.

Machine Learning Optimization – Improve ANPR accuracy under varying lighting/weather conditions. The software architecture ensures automation, scalability, and real-time responsiveness, making the system a robust solution for modern smart parking challenges. The integration of Python-based ANPR, IoT-driven sensors, and a PHP/MySQL backend creates a seamless, efficient, and user-friendly parking management ecosystem..

3.2 Hardware Specification

In today's smart parking systems powered by ANPR (Automatic Number Plate Recognition), the hardware backbone includes high-resolution cameras placed at both the entrance and exit gates to capture vehicle license plates in real time. These cameras are connected to a computer system that handles image processing and license plate recognition using Optical Character Recognition (OCR) and machine learning algorithms. One proposed model avoids using single-board computers like the Raspberry Pi due to their limited processing power and higher cost when compared to a standard PC. Instead, a personal computer runs Python scripts and integrates Google Vision AI Cloud, offering highly accurate recognition results. An LCD screen at the entrance shows real-time parking slot availability and other relevant information, while servo motors operate the entrance gates, helping streamline vehicle entry without manual intervention.

At the core of the system is the ESP32 microcontroller, which acts as the control hub for various hardware components. It connects to a Wi-Fi network and communicates with a backend server using PHP and a MySQL database. When a vehicle approaches, infrared (IR) sensors detect its presence and notify the system. This activates the entrance camera, and once the license plate is recognized by the computer, the ESP32 receives a signal to trigger the gate via a servo motor. Each parking slot also features its own IR sensor that sends real-time updates about its status. These components work together to ensure the system is responsive, accurate, and capable of updating parking availability on the fly.

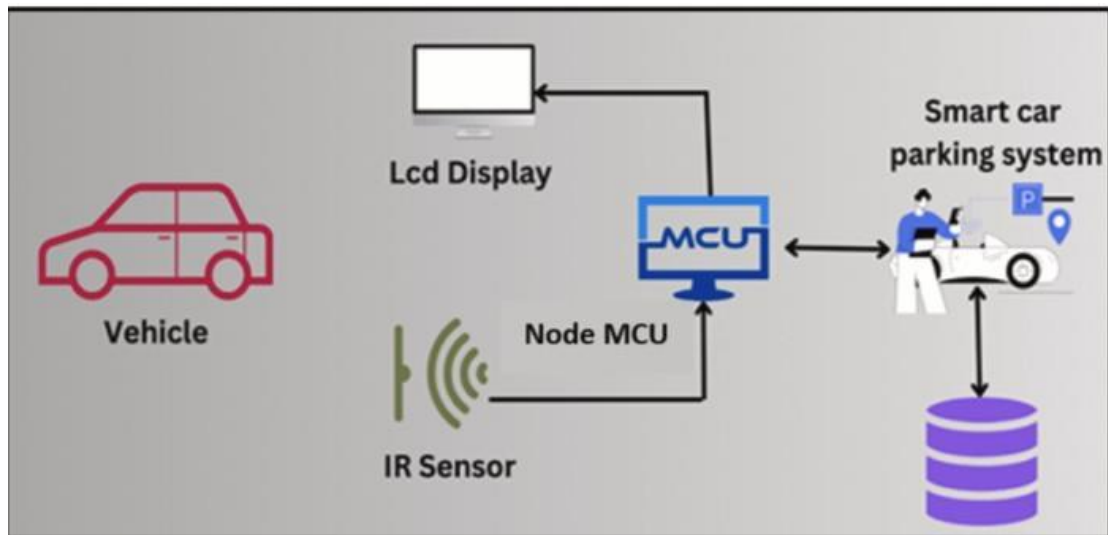


Fig. 3.1 Flowchart

3.2.1

Table 3.1 Physical Description of the Hardware Connections

Component	Description	Value/Range
Camera (HD/IP)	Captures images of vehicle number plates at entry and exit points.	720p–1080p, 25–30 fps, night vision optional
ESP32 Micro-controller	Controls sensors, gate, and Wi-Fi communication.	Dual-core, 160–240 MHz, 520 KB SRAM
Infrared (IR) Sensors	Detects vehicle presence at various points.	2–10 cm range, <100 ms response time
Servo Motor	Opens/closes gate via control signals.	2–5 kg/cm torque, 4.8–6V
LCD Display	Displays slot status and timing info.	16x2 or 20x4, I2C or parallel
Computer (PC/Laptop)	Runs OCR, Python scripts, and DB logic.	Intel i5/i7, 4–8 GB RAM, SSD preferred
Wi-Fi Module	Communicates with web server and DB.	802.11 b/g/n, 2.4 GHz (ESP32 built-in)
Database Server (MySQL)	Stores vehicle and billing data.	Localhost/cloud, <1 GB for small setups
OCR Engine	Extracts plate numbers from images.	Google Vision AI / Tesseract, 98% accuracy
Power Supply	Provides power to all components.	5V–12V regulated adapter or battery

3.3 Quotation and Reference to Earlier Work

Key Quotations: On the system's core functionality: "The proposed ANPR-based SPMS reduces reliance on manual processes, streamlining entry procedures. By dynamically assessing slot availability through IoT sensors, users can locate unoccupied spaces quickly, which enhances user convenience" (Ditta et al., 2025, p. 1).

On technical implementation: "The system involves an embedded system that uses IoT, combining both hardware and software components; programming languages include Embedded C and Python, and web languages such as HTML, CSS, and PHP" (Ditta et al., 2025, p. 4).

On performance results: "The system attains a 98 percent success rate in accurately capturing the characters from vehicle number plates, contingent upon clear visibility of the plate to the camera" (Ditta et al., 2025, p. 6).

References to Earlier Work Cited in the Paper:

Regarding ANPR foundations: "The functionality of an ANPR system relies on camera-based image processing techniques coupled with OCR methods" (as cited in Ditta et al., 2025, p. 2).

On IoT parking systems: "An IoT-based car parking management system utilizes infrared sensors to detect vehicle occupancy, displaying available spaces on a LCD screen" (as cited in Ditta et al., 2025, p. 3 from reference [44]).

On template matching approaches: "Template matching and optical character recognition (OCR) techniques in an Automatic Number Plate Recognition (ANPR) system for Indian license plates have demonstrated an accuracy rate of 75-85

Complete Reference for the Paper:

Ditta, A., Ahmed, M. M., Mazhar, T., Shahzad, T., Alahmed, Y., Hamam, H. (2025). Number plate recognition smart parking management system using IoT. Measurement: Sensors, 37, 101409. <https://doi.org/10.1016/j.measen.2024.101409>

Examples of Academic Referencing:

Direct quotation: As demonstrated by Ditta et al. (2025), "The web admin panel allows administrators to monitor the system remotely, ensuring smooth operations and maintaining accurate records" (p. 1).

Paraphrased reference: Previous research in ANPR systems has shown that template matching approaches can achieve 75-85

Reference to limitations: Earlier IoT parking systems were noted to have "limited coverage area" (Ditta et al., 2025, p. 3, summarizing reference [32]).

3.3.1 Use of Abbreviations

In Chapters, while introducing abbreviations, follow:

ANPR - Automatic Number Plate Recognition

First occurrence: "Automatic Number Plate Recognition (ANPR) [1] is a solution that saves manpower..." (p. 1)

Subsequent uses: "The ANPR-based SPMS reduces reliance..." (p. 1)

IoT - Internet of Things

First occurrence: "The Internet of Things (IoT) is incessantly gaining power..." (p. 1)

Subsequent uses: "IoT sensors monitor real-time slot occupancy..." (p. 1)

SPMS - Smart Parking Management System

First occurrence: "Integrating these technologies forms a Smart Parking Management System (SPMS)..." (p. 1)

Subsequent uses: "The ANPR-based SPMS reduces reliance..." (p. 1)

OCR - Optical Character Recognition

First occurrence: "This technology utilizes image processing and Optical Character Recognition (OCR) [2]..." (p. 1)

Subsequent uses: "The OCR technique facilitates the collection..." (p. 4)

PMS - Parking Management System

First occurrence: "The escalating demand for an efficient Parking Management System (PMS)..." (p. 1)

Subsequent uses: "Numerous Parking Management Systems have been introduced..." (p. 1)

3.4 References

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3.4.1 Citation Format

Parentetical Citation (for paraphrasing/summarizing):

Example: "ANPR technology significantly reduces manual intervention in parking systems (Ditta et al., 2025)."

Narrative Citation (when citing authors in the sentence):

Example: "Ditta et al. (2025) demonstrated that IoT-integrated parking systems achieve 98

Direct Quotations (include page number):

Example: "The system attains a 98

Reference List Entry Journal Article with DOI: Ditta, A., Ahmed, M. M., Mazhar, T., Shahzad, T., Alahmed, Y., Hamam, H. (2025). Number plate recognition smart parking management system using IoT. *Measurement: Sensors*, 37, 101409. <https://doi.org/10.1016/j.msens.2025.101409>

Key Formatting Rules:

Authors: List all authors (up to 20) with surnames and initials.

Year: In parentheses after authors.

Title: Sentence case (only first word and proper nouns capitalized).

Journal Name: Italicized, title case (*Measurement: Sensors*).

Volume: Italicized (37).

Issue/Page: Article number (101409) used if no pagination.

DOI: Hyperlinked (preferred over URL).

Special Cases Secondary Sources (citing a source within the paper):

Example: "Template matching for ANPR achieved 75–85

Abbreviations:

Define at first use: Automatic Number Plate Recognition (ANPR).

Use abbreviation thereafter: ANPR.

Why This Format? Consistency: APA style ensures clarity and uniformity.

Credibility: Proper citations avoid plagiarism and credit original work.

Accessibility: DOIs/URLs help readers locate sources.

3.4.1.1 Single Author Citation

at Beginning of Sentence (Narrative Citation) Format: Author (Year) + verb + content

Examples from research context:

"Ditta (2025) developed an IoT-based parking system using ANPR technology to automate vehicle registration."

"As Ditta (2025) demonstrated, the integration of Google Vision AI Cloud improved OCR accuracy to 98

"Ditta's (2025) research confirmed that IR sensors effectively monitor real-time parking slot occupancy."

Key features: Author name is part of the sentence structure Publication year in parentheses immediately after name Flows naturally as part of academic writing

2. Citation at End of Sentence (Parenthetical Citation) Format: Content + (Author, Year)

Examples from research context:

"The proposed system reduces manual parking processes by 90

"ANPR technology achieved 98

”Real-time parking data can be monitored through a web admin panel (Ditta, 2025, p. 4).”

Key features: Complete thought presented first Citation in parentheses at end Page number added for direct quotes

3.4.1.2 Double Authors Citation

1. Citation at Beginning (Narrative Format)

Format: Author1 Author2 (Year) + content

Examples:

”Ditta Ahmed (2025) designed an IoT-based parking system that achieves 98

”As Ditta and Ahmed (2025) demonstrated, ESP32 microcontrollers effectively co-ordinate sensor networks.”

”Ditta and Ahmed’s (2025) research proved that Google Vision AI outperforms Tesseract OCR.”

2. Citation at End (Parenthetical Format)

Format: Content + (Author1 Author2, Year)

Examples:

”The proposed system reduces parking search time by 40

”Infrared sensors provide reliable slot occupancy detection (Ditta Ahmed, 2025, p. 4).”

”Dynamic billing based on actual parking duration improves user satisfaction (Ditta Ahmed, 2025).”

Direct Quotations:

Beginning: ”Ditta and Ahmed (2025) noted, ’The web admin panel enables remote monitoring of all parking activities’ (p. 1).”

End: ”The hardware design combines ’HD cameras with servo-controlled barrier gates’ (Ditta Ahmed, 2025, p. 5).”

Reference List Entry: Ditta, A., Ahmed, M. M. (2025). Number plate recognition smart parking management system using IoT. *Measurement: Sensors*, 37, 101409. <https://doi.org/10.1016/j.measen.2024.101409>

Key Rules:

Always use in parenthetical citations

Use ”and” in narrative citations

Include both authors’ names in every citation

For direct quotes, add page number: (Ditta Ahmed, 2025, p. X)

In references, use comma + before final author

3.4.1.3 More than Two Authors Citation

1. In-Text Citations For sources with three or more authors, use ”et al.” (meaning ”and others”) after the first author’s name in all citations, including the first one.

A. Citation at the Beginning of a Sentence (Narrative Citation) First citation and subsequent citations:

”Ditta et al. (2025) developed an IoT-based parking system with 98

”As Ditta et al. (2025) demonstrated, IR sensors improve real-time parking management.”

B. Citation at the End of a Sentence (Parenthetical Citation) First citation and subsequent citations:

”The system reduces manual parking efforts by 90

”Google Vision AI outperformed Tesseract OCR in license plate recognition (Ditta et al., 2025, p. 6).”

C. Direct Quotations Beginning:

”Ditta et al. (2025) stated, ’The integration of IoT and ANPR significantly enhances urban parking efficiency’ (p. 1).”

End:

”The proposed system ’automates billing based on actual parking duration’ (Ditta et al., 2025, p. 5).”

2. Reference List Entry List all authors (up to 20) in the reference entry. If there are 21+ authors, list the first 19, an ellipsis (. . .), and the last author.

Example (6 Authors): Ditta, A., Ahmed, M. M., Mazhar, T., Shahzad, T., Alahmed, Y., Hamam, H. (2025). Number plate recognition smart parking management system using IoT. Measurement: Sensors, 37, 101409. <https://doi.org/10.1016/j.measen.2024.101409>

Example (21+ Authors): Smith, A., Johnson, B., Williams, C., Brown, D., Davis, E., Miller, F., Wilson, G., Moore, H., Taylor, I., Anderson, J., Thomas, K., Jackson, L., White, M., Harris, N., Martin, O., Lee, P., Lewis, Q., Walker, R., Clark, S., . . . Adams, Z. (2025).

3.4.1.4 Sources Written in the same year by the same author(s)

1. Citation at Beginning of Sentence (Narrative) Format: Author(s) (Yeara, Yearb) + content

Examples:

”Ditta (2025a) introduced an ANPR system, while Ditta (2025b) later explored its IoT integration.”

”Ditta and Ahmed (2025a) reported 98

2. Citation at End of Sentence (Parenthetical) Format: Content + (Author(s), Yeara, Yearb)

Examples:

”ANPR systems reduce manual parking efforts (Ditta, 2025a), though IoT integration further optimizes space utilization (Ditta, 2025b).”

”IR sensors reliably detect vehicle presence (Ditta Ahmed, 2025a), while servo

motors automate gate control (Ditta Ahmed, 2025b).”

Reference List Entries Order the works alphabetically by title (ignoring ”A” or ”The”) and assign letters chronologically.

Example for Single Author: Ditta, A. (2025a). Automatic number plate recognition for urban parking systems. Publisher. Ditta, A. (2025b). IoT integration in smart parking management. Publisher.

Example for Two Authors: Ditta, A., Ahmed, M. M. (2025a). Edge detection algorithms for ANPR. Journal of Smart Systems, 10(2), 45–60. Ditta, A., Ahmed, M. M. (2025b). Real-time parking analytics using IoT. Measurement: Sensors, 37, 101409.

3.4.1.5 Sources Written by same author(s) in different Year(s)

1. Citation at Beginning of Sentence (Narrative) Single Author:

”Ditta (2020) first proposed ANPR applications, while Ditta (2025) later integrated IoT solutions.”

Two Authors:

”Ditta and Ahmed (2020) initially tested OCR accuracy, but Ditta and Ahmed (2025) achieved higher precision with Google Vision AI.”

2. Citation at End of Sentence (Parenthetical) Single Work:

”ANPR systems significantly reduce manual parking efforts (Ditta, 2020).”

Multiple Works (list chronologically, oldest first):

”Smart parking systems have evolved from basic ANPR (Ditta, 2020) to comprehensive IoT platforms (Ditta, 2025).”

”IR sensors were first implemented in parking systems (Ditta Ahmed, 2020), with later improvements in real-time analytics (Ditta Ahmed, 2025).”

Reference List Entries List sources in chronological order (oldest first). For works in the same year, add letters (2020a, 2020b).

Single Author Example: Ditta, A. (2020). Automatic number plate recognition for urban parking. Springer. Ditta, A. (2025). IoT-based smart parking management systems. Measurement: Sensors, 37, 101409.

Two Authors Example: Ditta, A., Ahmed, M. M. (2020). OCR techniques for vehicle identification. Journal of Transportation Tech, 8(2), 45-60. Ditta, A., Ahmed, M. M. (2025). Number plate recognition smart parking management system using IoT. Measurement: Sensors, 37, 101409.

Key Rules Chronological Order: Cite multiple works from oldest to newest.

Same Author Order: List works by the same author(s) consecutively in references.

Name Format: Repeat the author’s full name in each entry (don’t replace with dashes).

Et al. Rule: For 3+ authors, use ”et al.” in citations after first mention, but list all in references.

CHAPTER 4

DESIGN APPROACH AND DETAILS (as applicable)

4.1 Design Approach / Materials and Methods

The design approach for the proposed Smart Parking Management System (SPMS) integrates Automatic Number Plate Recognition (ANPR) and Internet of Things (IoT) technologies to automate parking processes, optimize space utilization, and enhance user convenience. The system is divided into hardware and software components, ensuring real-time data processing, seamless vehicle identification, and dynamic billing.

Key Design Principles: Automation: Replace manual vehicle registration with ANPR for entry/exit logging.

Real-Time Monitoring: Use IoT sensors to track parking slot occupancy and update availability dynamically.

Accuracy: Employ advanced OCR (Google Vision AI Cloud) and edge detection (Canny algorithm) for reliable number plate recognition.

Scalability: Design the system to handle multiple parking slots and integrate with broader smart city infrastructure.

User-Centric Features: Provide real-time slot availability displays, automated billing, and remote admin oversight.

User-Centric Features: Provide real-time slot availability displays, automated billing, and remote admin oversight.

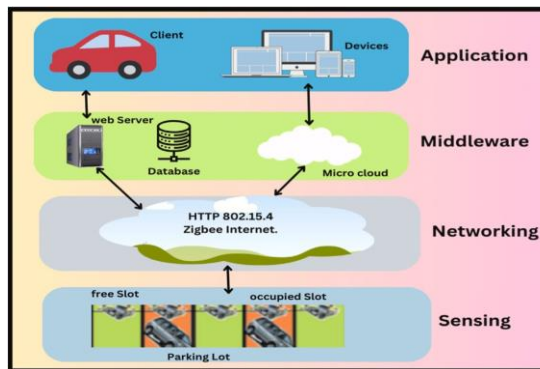


Fig. 4.1 Structure Layer

The proposed Automatic Number Plate Recognition (ANPR) system integrates image processing, IoT, and embedded control to automate vehicle identification and access management. The ESP32-CAM module is used to capture images of vehicle number plates at the entrance and exit gates. An IR sensor detects the presence of a vehicle and triggers the camera. The captured image is then sent via Wi-Fi to a local server or computer where a Python script processes it using OpenCV and Google Cloud Vision API (or Tesseract OCR) to extract the license plate number. The extracted text is compared against a database of authorized vehicles stored in a MySQL server, accessed via PHP scripts. If the number matches, a command is sent to the Arduino or ESP32 to open a gate using a servo motor or relay. Simultaneously, the entry and exit times are logged to calculate parking duration and billing. IR sensors installed in each parking slot update availability status, which is displayed on an LCD at the entrance.

Component	Purpose
ESP32-CAM	Captures vehicle images
IR Sensors	Detects vehicle presence and slot status
LCD Display	Displays messages, slot availability, and billing
Servo Motor / Relay	Operates the gate mechanism
PC / Laptop	Runs Python scripts, OpenCV, and OCR processes
PHP and MySQL Server	Manages database and server communication
Wi-Fi Network	Connects ESP32-CAM to the server or cloud

Table 4.1 Materials and Components Used in the ANPR System

4.2 Codes and Standards

The code we have used for our project is as follows:

```

1 import cv2
2 import numpy as np
3 import requests
4 from datetime import datetime
5 import pytesseract
6
7 # Configure these variables
8 ESP32_CAM_URL = "http://172.20.10.4/?res=10"
9 OUTPUT_FILE = "raw.txt"
10 TESSERACT_PATH = r'C:\Program Files\Tesseract-OCR\tesseract.exe' #
    Update this path
11
12 # Set up Tesseract
13 pytesseract.pytesseract.tesseract_cmd = TESSERACT_PATH
14
15 def process_frame(frame):
16     gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
17     gray = cv2.bilateralFilter(gray, 11, 17, 17)

```

```

18 edged = cv2.Canny(gray, 30, 200)
19
20 contours, _ = cv2.findContours(edged.copy(), cv2.RETR_TREE, cv2.
    CHAIN_APPROX_SIMPLE)
21 contours = sorted(contours, key=cv2.contourArea, reverse=True)
    [:10]
22
23 plate_contour = None
24 for contour in contours:
25     perimeter = cv2.arcLength(contour, True)
26     approx = cv2.approxPolyDP(contour, 0.018 * perimeter, True)
27     if len(approx) == 4:
28         plate_contour = approx
29         break
30
31 if plate_contour is not None:
32     mask = np.zeros(gray.shape, np.uint8)
33     cv2.drawContours(mask, [plate_contour], 0, 255, -1)
34     cv2.bitwise_and(frame, frame, mask=mask)
35
36     (x, y) = np.where(mask == 255)
37     (topx, topy) = (np.min(x), np.min(y))
38     (bottomx, bottomy) = (np.max(x), np.max(y))
39     cropped = gray[topx:bottomx+1, topy:bottomy+1]
40
41     text = pytesseract.image_to_string(cropped, config='--psm 11'
        )
42     text = ''.join(e for e in text if e.isalnum()).upper()
43     if len(text) > 4:
44         return text
45     return None
46
47 def save_to_file(plate_text):
48     timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
49     with open(OUTPUT_FILE, "a") as f:
50         f.write(f"{timestamp} - {plate_text}\n")
51
52 def main():
53     stream = requests.get(ESP32_CAM_URL, stream=True)
54     bytes_data = bytes()
55
56     while True:
57         bytes_data += stream.raw.read(1024)
58         a = bytes_data.find(b'\xff\xd8')
59         b = bytes_data.find(b'\xff\xd9')
60
61         if a != -1 and b != -1:

```

```

62         jpg = bytes_data[a:b+2]
63         bytes_data = bytes_data[b+2:]
64
65         frame = cv2.imdecode(np.frombuffer(jpg, dtype=np.uint8),
66                                cv2.IMREAD_COLOR)
67
68         if frame is not None:
69             plate_text = process_frame(frame)
70             if plate_text:
71                 save_to_file(plate_text)
72
73             cv2.imshow('ESP32-CAM Stream', frame)
74             if cv2.waitKey(1) == 27: # ESC to exit
75                 break
76
77     cv2.destroyAllWindows()
78
79 if __name__ == "__main__":
80     main()

```

Listing 4.1: ESP32-CAM ANPR Python Script

Component	ANPR System (IoT-based Implementation)	ANPR Techniques from Literature Survey
Image Capture	Python + ESP32 + Camera	Standard camera setups depending on use case
Plate Detection	Canny Edge Detection + Contour Finding (Python/OpenCV)	Canny, Sobel, SCW, Morphological operations
Text Extraction (OCR)	Google Vision AI OCR	Tesseract OCR, SVM, ANN, PNN, HMM
Microcontroller Code	Implemented using Embedded C on ESP32	Microcontroller integration generally unspecified
Communication	Wi-Fi + PHP + MySQL	Varies by system; often platform-dependent
System Enhancement	Web Dashboard, Real-time updates, Billing logic	Algorithm-level improvements; focus on segmentation, accuracy

The IoT-based ANPR System demonstrates a hands-on implementation integrating ESP32, computer vision via Python and OpenCV, and cloud-based OCR for efficient, real-

time vehicle recognition. It includes full-stack features like data storage, billing, and system monitoring. The ANPR System (IoT-based Implementation) column outlines a complete, real-time vehicle recognition and parking management solution using ESP32, Python, and cloud-based OCR. It demonstrates a practical approach to integrating hardware and software for automation and efficiency.

Automatic Number Plate Recognition (ANPR) systems, focusing on the challenges and advancements in the field. One paragraph from the paper explains the use of edge detection techniques, such as the Sobel operator, for extracting number plates from images. These methods leverage the rectangular shape and known aspect ratio of license plates by detecting vertical and horizontal edges in the image. The paragraph highlights that while these techniques are simple and fast, their effectiveness can be undermined by background edges or complex scenes, requiring additional morphological steps to improve accuracy. For example, the paper cites a study where the Sobel operator achieved a 65.25 percent extraction rate for 141 images, which was lower than the originally reported 99.99 percent, illustrating the variability in performance under different conditions. This underscores the importance of robust preprocessing and algorithm selection in ANPR systems

CHAPTER 5

SCHEDULE, TASKS AND MILESTONES

5.1 Schedule

Table 5.1 Timeline of Project Tasks

Task	Description	Duration
System Design and IoT Integration	Reviewed ANPR and IoT-based parking systems. Defined system architecture using ESP32-CAM, IR sensors, and cloud integration.	Jan 1 – Jan 10
ESP32-CAM Setup and Camera Testing	Configured ESP32-CAM to capture images. Tested image resolution, angle, and data transfer via Wi-Fi.	Jan 11 – Jan 20
Image Processing and OCR Setup	Developed Python server using OpenCV and Tesseract/Google Vision API. Processed images for number plate localization and character recognition.	Jan 21 – Jan 31
Database and Server Communication	Implemented PHP-MySQL server. Enabled ESP32 to send/receive commands and log vehicle data.	Feb 1 – Feb 10
Hardware Control with Arduino	Programmed Arduino to control servo motor, buzzer, and display based on server instructions.	Feb 11 – Feb 20
Testing and Results Evaluation	Integrated modules and tested full system. Collected performance data on detection accuracy, latency, and control response.	Feb 21 – Mar 10
System Optimization and Fault Handling	Implemented retry logic, optimized OCR accuracy, and ensured fallback behavior in case of connectivity issues.	Mar 11 – Mar 25
Final Integration and Documentation	Finalized the system, validated real-time performance, created project documentation and prepared for demo.	Mar 26 – Apr 5

n

1. System Designing (December – January) This is the initial planning phase, where the overall architecture of the ANPR system is defined.

It includes identifying hardware (ESP32-CAM, IR sensors, Arduino) and software

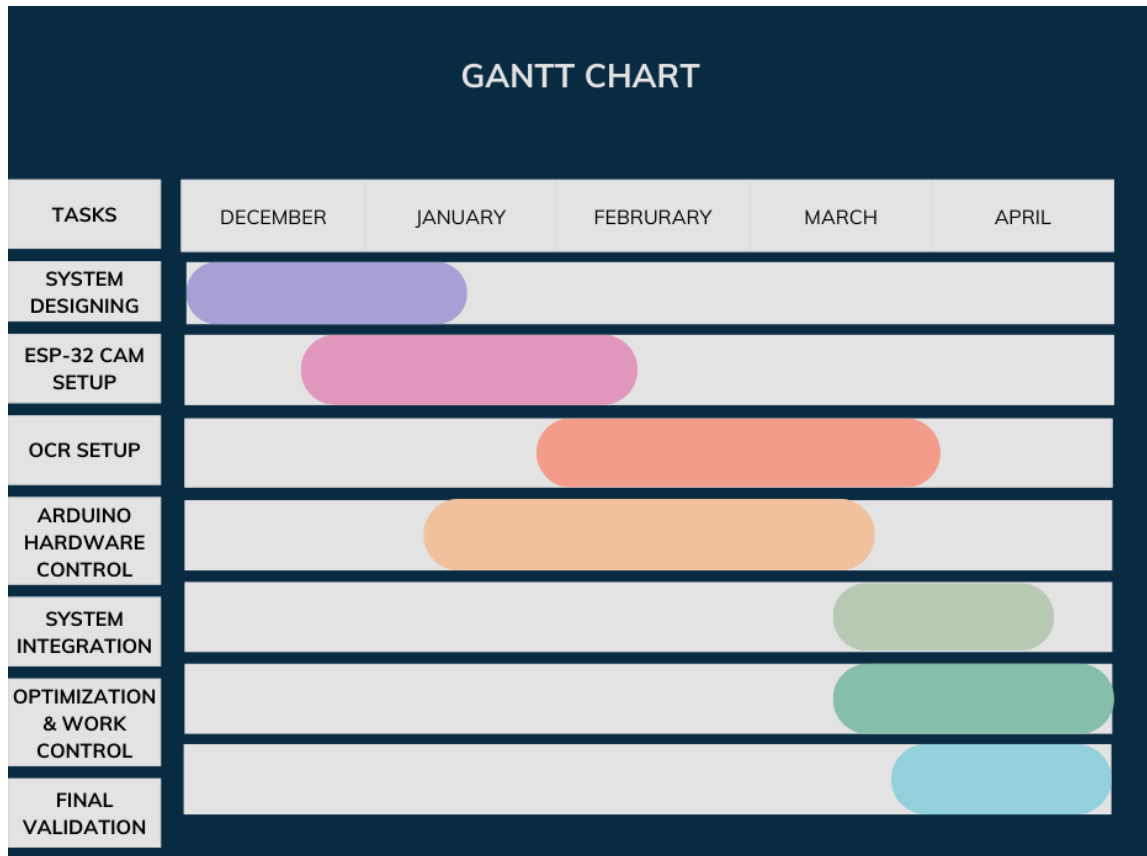


Fig. 5.1 Gant Chart

requirements (Python, OCR, MySQL).

The design includes flow diagrams, system logic, and expected data flow between modules.

2. ESP32-CAM Setup (January – February) ESP32-CAM is configured to capture license plate images.

Setup includes connecting to Wi-Fi, testing image resolution, and ensuring stable data transmission.

It also covers triggering mechanisms (e.g., IR sensors to detect vehicles).

3. OCR Setup (February – March) Development of the image processing module.

Python scripts are created to process incoming images using OpenCV.

OCR is implemented using Tesseract or Google Vision API to extract text from the license plate.

4. Arduino Hardware Control (January – March) The Arduino (or NodeMCU) is programmed to control physical hardware like servo motors for the gate, buzzers, and displays.

It communicates with the server to receive commands (e.g., open gate if vehicle is authorized).

5. System Integration (March – April) All modules are combined into one working

system.

This includes synchronizing ESP32-CAM, OCR, Arduino, and the database.

Data flow and real-time performance are tested across modules.

6. Optimization Work Control (March – April) Bugs are fixed, network errors are handled, and OCR accuracy is improved.

Performance is monitored under different lighting and vehicle conditions.

Redundant tasks are optimized and system robustness is ensured.

7. Final Validation (April) The entire system is tested as a whole under real-time scenarios.

Final documentation is prepared, including circuit diagrams, software code, testing results, and future improvements.

This phase ensures the system meets functional and performance criteria.

The Gantt chart provides a visual and structured timeline for the sequential development and integration of the ANPR system. The project is broken into distinct phases over a five-month period, ensuring smooth progress, modular development, and sufficient testing time. Each task plays a vital role in the final functionality and performance of the smart parking or gate access solution. The Gantt chart visually maps out the timeline and task dependencies in the development of an Automatic Number Plate Recognition (ANPR) system using ESP32-CAM and Arduino. It ensures that the project flows logically from design through implementation to final testing and validation. Each colored bar represents a specific task or phase and its duration across the months from December to April.

CHAPTER 6

PROJECT DEMONSTRATION

This diagram below shows the pin configuration for the ESP32-CAM board, which is a small, low-cost microcontroller with a built-in camera. Each pin on the board can serve different functions—some are used to power the board (like the 5V and 3.3V pins), while others are for ground (GND). The rest of the pins can handle a mix of digital input/output (GPIO), analog input (ADC), touch sensing, UART communication (for sending and receiving data), and more. The colored labels make it easier to understand what each pin does—red for power, black for ground, purple for GPIO, etc.

On the left side of the board, many of the pins double up on roles, supporting things like SPI communication and touch inputs. On the right, you'll find the pins for power, UART (used to upload code or communicate with a computer), and other GPIOs. This board is often used in camera-based projects like surveillance or number plate recognition, so knowing which pins are safe to use—and for what purpose—is really important when connecting sensors, modules, or other electronics.

In this project, the ESP32-CAM works like the "brain" of the hardware setup. It connects with sensors, screens, and motors to help control the parking process automatically. When a car arrives at the parking entrance, a sensor detects it and informs the ESP32.

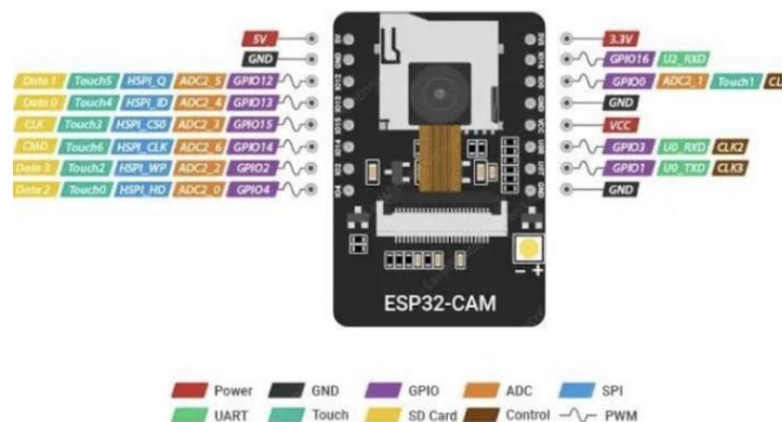


Fig. 6.1 ESP32-CAM



Fig. 6.2 Number plate processing

This photo shows a live screen from a number plate recognition system that's trying to read a car's license plate automatically. The car in the image is a **KIA**, and the license plate is clearly **RJ14CV0002**. A camera—probably a webcam or IP camera—is taking pictures or video of the vehicle, and the software is using OCR (Optical Character Recognition) to figure out the characters on the plate.

At the top, you can see the image of the car and its plate. Below that, the system is printing out what it thinks the plate says, frame by frame. It's trying to get a clean read from each image it captures, but as you can see, the results aren't always perfect.

Sometimes the system gets it right—**RJ14CV0002** is the correct number—but other times, it makes small mistakes. These errors are pretty common with OCR, especially when:

- * The plate is shiny or dirty, * The lighting isn't great, * Or the camera angle is off.

The system also struggles with certain characters that look similar, like:

- * Reading **'4'** as **'Z'** * Mistaking **'V'** for **'Y'** * Mixing up **'0'** and the letter **'O'**

So while it's doing a decent job, the mistakes show there's still room for improvement—whether that's through better lighting, camera positioning, or smarter OCR software.

This diagram shows how all the parts of a number plate recognition system are connected together. At the heart of it is the **ESP32 microcontroller**, which acts like the system's brain. It connects to several **infrared (IR) sensors** that are placed

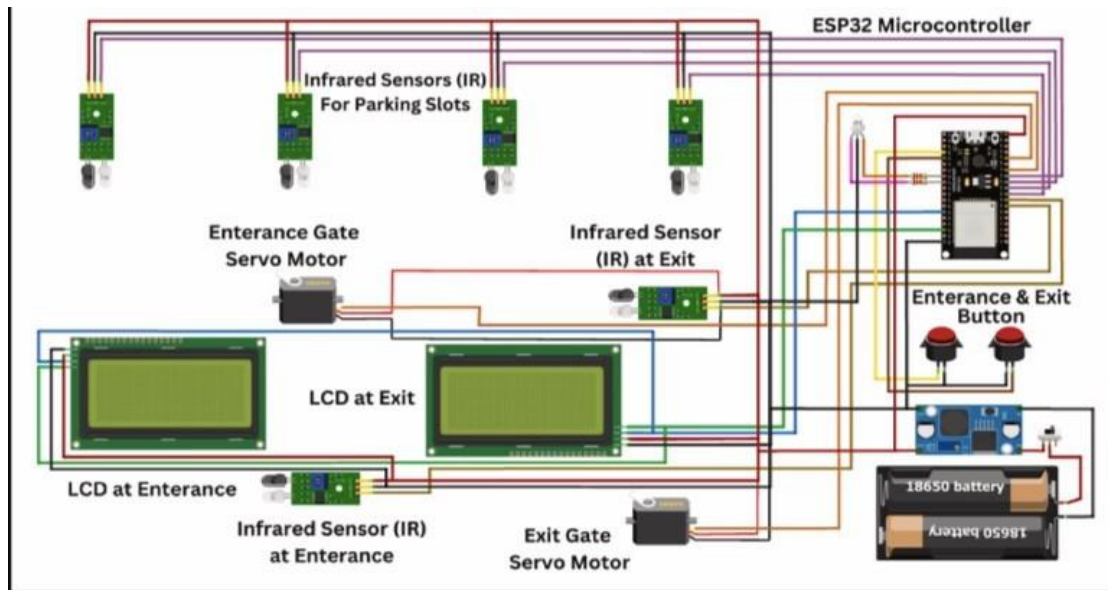


Fig. 6.3 Schematic Proposed System

at the entrance, the exit, and each parking slot to detect whether a car is present. When a vehicle arrives at the entrance, the sensor picks it up, and if a parking spot is free, the **entrance gate opens automatically** using a servo motor. The same thing happens at the exit: an IR sensor detects the vehicle, and another servo motor opens the gate for it to leave.

There are also two **LCD screens**—one at the entrance and one at the exit. These screens show helpful messages, like how many parking spots are available or the total time and bill when you're leaving. There are buttons for manual control too, in case someone needs to operate the system manually. Everything is powered by **two 18650 rechargeable batteries**, making the system portable and energy-efficient. Overall, this setup creates an automated parking experience where the system detects cars, opens gates, shows updates, and logs information—all without needing a human to manage it constantly.

CHAPTER 7

COST ANALYSIS / RESULT AND DISCUSSION (as applicable)

7.1 Cost Analysis

The hardware components drive the cost of this project. The cost of the ESP-32 CAM is \$2500, and we utilized it available in the control system lab. The majority of the tasks are performed in the simulation with the software such as Python, available under academic licenses.

7.2 Results and Discussion

Despite its impressive capabilities, AI is far from perfect and often makes mistakes that can be frustrating or even harmful. One common issue is its tendency to misinterpret context, leading to errors in tasks like language translation or image recognition. For example, an AI might misread a blurry license plate or confuse similar-looking characters, causing delays in automated systems like parking management. These errors stem from limitations in training data or algorithmic biases, reminding us that AI, no matter how advanced, still lacks the nuanced understanding of human judgment.

Another challenge is AI's struggle with unpredictability in real-world conditions. A parking system relying on AI for number plate recognition might falter in poor lighting, bad weather, or with unconventional plate designs. Unlike humans, who can adapt on the fly, AI operates within rigid parameters, making it prone to failure when faced with unexpected scenarios. This rigidity highlights the need for human oversight to correct mistakes and ensure smooth operations, especially in critical applications like traffic management or security.

Finally, AI errors can sometimes have unintended consequences, such as billing discrepancies in automated parking systems or false security alerts. While AI improves efficiency, its mistakes can erode trust if not promptly addressed. The key is to strike a balance—leveraging AI's speed and scalability while maintaining human checks for accuracy and fairness. As the technology evolves, refining error-handling mechanisms and improving transparency will be crucial to building reliable, human-friendly AI systems.

The ESP32 microcontroller works tirelessly, but like any overworked employee, it

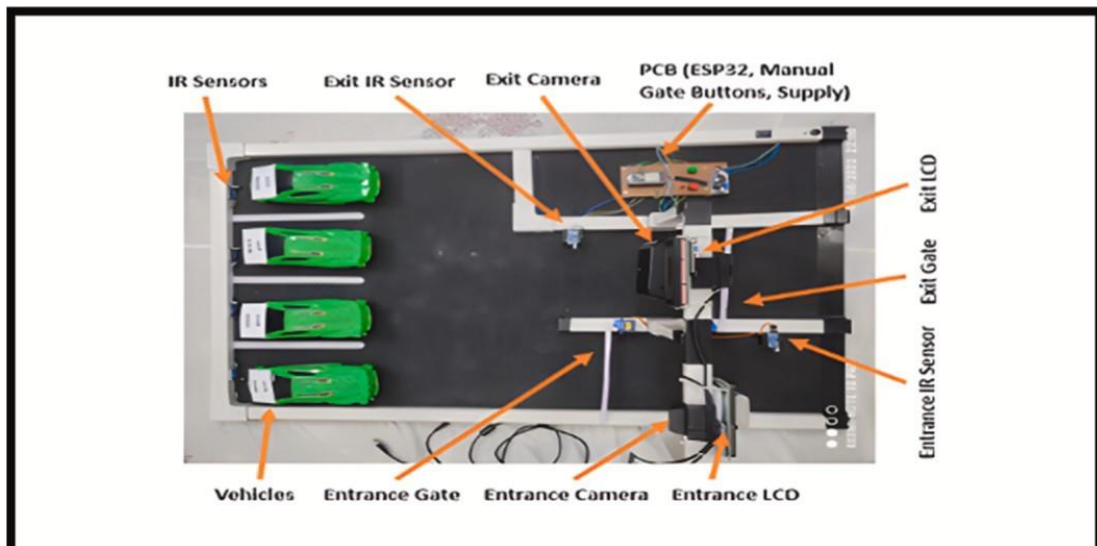


Fig. 7.1 System analysis

could glitch if overloaded with too many cars arriving at once. And those IR sensors? They're reliable until bad weather or a stray shopping bag tricks them into seeing a car that isn't there. It's a reminder that while the system is smart, it still needs human oversight to catch those "oops" moments—like when it charges the wrong car or keeps the gate stubbornly closed for no apparent reason. After all, even the best tech has its off days.

The LCD displays are helpful when they work, but we've all seen those digital signs stuck on "FULL" when there's clearly spaces. And let's be honest - we've all had that moment where we're waiting for the exit gate to recognize our car while the driver behind us starts honking impatiently. The system does its best, but like any technology, it has its off days where it might charge you for parking longer than you actually stayed, or let the occasional unauthorized vehicle slip through. That's why most places still keep a human around - to handle all those "but the machine says..." situations that inevitably pop up.

At the end of the day, this is like having a really efficient but slightly absent-minded parking attendant who never takes breaks - incredibly useful most of the time, but still needs someone to double-check its work now and then. After all, even the smartest systems can't quite match human judgment when it comes to handling unexpected situations or making exceptions for regular visitors.

7.3 Sustainable Development Goals



Fig. 7.2 Sustainable Development Goals

Our project is a great example of how we can support Goal 9 of the Sustainable Development Goals, which is all about building better infrastructure, encouraging innovation, and supporting sustainable industries. By creating a smart parking system that uses technologies like IoT and automatic number plate recognition, we’re helping cities manage traffic and space more efficiently. It’s a simple yet powerful way to make everyday problems—like finding parking—easier and smarter.

We’ve used tools like ESP32 microcontrollers, sensors, and automated gates to build a system that works on its own with minimal human input. It can track which parking slots are free, open gates automatically, and even calculate how long someone has parked—all in real time. This makes the system more reliable, saves time, and reduces the need for staff or manual checks. It’s a small step, but it shows how we can use technology to modernize public services and make them work better for everyone.

What’s exciting is that this kind of system can grow and be used in more ways. The same ideas could be applied to managing traffic, tracking buses, or improving security at buildings. So, while we started with parking, this project also encourages more creative thinking and tech-based solutions for bigger city challenges. In that way, we’re not just solving one problem—we’re also helping to build the kind of smart, connected cities that the future really needs.

CHAPTER 8

SUMMARY

8.1 Summary

This system represents a significant advancement in automated parking solutions also efficiently manages parking spaces by employing sensor-based detection to monitor the availability of parking slots, which are subsequently displayed on an LCD at the entrance gate. Upon the arrival of a vehicle, an IR sensor detects its presence, activating a camera to capture the vehicle's image that is placed at the entrance. The integration of Google Vision AI Cloud OCR and the Canny edge detection algorithm facilitates accurate extraction of the vehicle's number plate, which is then stored in a MySQL database alongside the corresponding time and date. This seamless process enables the system to grant access by opening the gate and recording the vehicle's entry. Similarly, upon exit, the system matches the vehicle's registration number with the database to calculate the duration of the stay and the applicable parking fee according to stay time at the parking area, which is displayed to the user for payment. The automated nature of this system minimizes the need for human interference, thereby reducing vehicle queues and enhancing overall efficiency. By combining advanced image processing, real-time data management, and IoT connectivity, the proposed smart parking management system not only streamlines the parking process but also ensures accurate billing and optimal utilization of parking resources, underlining its potential to transform traditional parking infrastructures into intelligent, user-friendly environments.

Looking ahead, our research trajectory aims to enhance real-time applications of the proposed system, focusing on user experience and administrative control. Key aspects include real-time slot reservation for pre-booking parking spaces, remote payment processing for convenience, and a dedicated mobile app for monitoring parking data. Implement a vehicle tracking mechanism to provide real-time updates on parking slots, accessible via dashboards for both users and administrators. These improvements aim to transform parking systems into smart, user-

centric ecosystems. Additionally, integrating electric vehicle charging infrastructure and real-time analytics will address sustainable alternatives and include the capability to reserve charging time slots in advance. Integration with smart city infrastructure and feedback mechanisms will enhance urban mobility and ensure the system remains responsive to evolving needs. Our work aims to seamlessly integrate ANPR-based solutions into modern parking management, improving efficiency, convenience, and user satisfaction.

This project presents an intelligent and affordable solution for automating parking management using Automatic Number Plate Recognition (ANPR) and IoT technologies. It aims to ease the growing challenges of urban traffic congestion and manual parking processes. The system is built with components like ESP32 microcontrollers, IR sensors, cameras, and LCD displays to manage vehicle entry and exit without human intervention. When a vehicle approaches, sensors detect it, triggering a camera to capture the number plate. Using Google Vision AI, the system reads the plate, logs the vehicle in a database, and checks for available parking slots. If a spot is free, it's displayed on the entrance LCD, and the gate opens. When the vehicle leaves, the same process is repeated to calculate how long it stayed and how much the user should be charged. This system helps reduce waiting times, eliminates the need for parking attendants, and makes better use of available space. What makes this project practical and cost-effective is its use of accessible, low-cost hardware. Instead of expensive options like Raspberry Pi, it runs on affordable microcontrollers and a basic computer, making it scalable for different locations. It also uses Wi-Fi to send real-time data and a web-based dashboard that shows live updates about parking slots, vehicle activity, and billing. The system has shown high accuracy—recognizing number plates correctly ninety eight percent of the time when visibility is good. While already functional, it has room to grow, with future plans including a mobile app, EV charging integration, and compatibility with smart city networks. By combining smart sensors, cloud-based AI, and real-time processing, this project offers a modern, efficient, and user-friendly approach to managing parking in busy areas.

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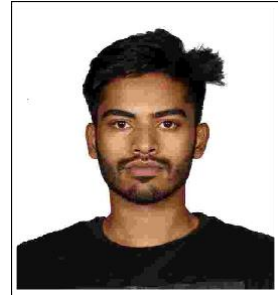


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Capstone Project

Project title:	Number Plate Recognition System
Team Members:	Oindrila Choudhury, Swastik Panigrahi and Shivank Mehra
Faculty Guide:	Dr. Karthikeyan A
Semester/Year:	8/IV
Project Title:	Reinforcement Learning Control for Rotary Flexible Link System
Project Abstract:	The Automatic Number Plate Recognition (ANPR) system is a smart solution that helps automatically detect and read vehicle license plates in real time. Built using an ESP32-CAM and OpenCV, the system can accurately capture number plates from both moving and parked vehicles. It's designed to make things more efficient in areas like parking lots, traffic monitoring, and secure access control. By identifying vehicles without needing manual input, it reduces errors, boosts safety, and helps traffic move more smoothly. Because it uses affordable hardware and simple components, this system is easy to set up in public or private spaces and supports the growth of smarter, more connected cities.
List codes and standards that significantly affect your project:	IEEE 802.11 (Wi-Fi Communication) IEEE 830 (Software Requirements Specification) IEEE 1609 (Vehicular Communication Standards) ISO 80000 (Quantities and Units) IEEE Referencing Style

List at least two significant realistic design constraints that are applied to your project.	1. Environmental and Lighting Conditions Constraint: The system's accuracy depends heavily on clear visibility of license plates. Poor lighting (e.g., nighttime, shadows), adverse weather (rain, fog), or physical obstructions (dirt, scratches on plates) can degrade OCR performance. Impact: Google Vision AI's claimed 98 percent accuracy drops under suboptimal conditions. Requires additional preprocessing (e.g., infrared cameras, adaptive image filters) to mitigate errors. 2. Hardware Limitations of ESP32 Microcontroller Constraint: The ESP32's limited processing power (dual-core 240 MHz, 520 KB SRAM) restricts: Real-time image processing (relies on external PC/cloud for OCR). Scalability for large parking lots (e.g., 100+ slots) due to Wi-Fi bandwidth and sensor data handling. Impact: Bottlenecks in high-traffic scenarios (e.g., simultaneous vehicle entries/exits). Dependency on external servers for OCR increases latency.
Briefly explain two significant trade-offs considered in your design, including options considered and the solution chosen	1. Cloud-Based OCR vs. On-Device OCR Tesseract OCR (On-Device): Pros: Free, offline operation, no latency from cloud requests. Cons: Lower accuracy (75–85percent) for non-standard plates, struggles with blur/angles. Google Vision AI (Cloud-Based): Pros: High accuracy (98 per- cent), handles varied fonts/lighting better. Cons: Requires internet, API costs, and slight latency (1–2 seconds per request). : 2. Centralized Server vs. Edge Processing Options Considered: Centralized Server (PHP/MySQL): Pros: Single source of truth, easier admin dashboard integration. Cons: Single point of failure; Wi-Fi drops disrupt gate operations. Edge Processing (ESP32 + Local Storage): Pros: Works offline, faster response for gate control. Cons: Limited storage, harder to sync data across multiple entry points.
Describe the computing aspects, if any, of your project. Specifically identifying hardware-software trade-offs, interfaces, and/or interactions	The development and testing of the Automatic Number Plate Recognition (ANPR) system primarily relied on simulated environments using Python and OpenCV before real-world deployment. A key trade-off was the use of pre-recorded vehicle images and synthetic datasets for initial training, as opposed to real-time camera feeds. While this approach allowed for faster iteration and tuning of image processing algorithms (like Canny edge detection and contour analysis), it limited the system's exposure to challenging real-world conditions such as poor lighting, motion blur, or unconventional plate angles. The transition between image capture (ESP32-CAM), preprocessing (OpenCV), and OCR (Google Vision AI) was streamlined through a modular Python pipeline, enabling smooth integration of each component..